

WRRRA M Integrated Upstream and Downstream Model

From finite source generation to common state energy currents
measurement records and gravitational response

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Evaluation summary and abstract

Verified inputs. We use frozen WRRRA Core 1.0 and Minimal Computation Cosmology MCC 2.3.2, the reviewed upstream, downstream and bridge studies, established constants and adopted observational calibration. Finite address support, internal states, species populations, physical responses, supply boundaries and clock choices are individually declared.

WRRRA transformation. Finite SOURCE addresses and composite relations are partitioned by ordered admission and return. Phenotype-conditioned addresses prepare states of a common internal Hamiltonian. Energy and currents are read from that same state, and the existing phenotype allocation is replaced by rest and excitation energies. Supplier and measurement-record changes are counted once, and pressure and homogeneous expansion follow from the volume derivative of the same energy. At an adopted proper-time spacing we execute post-measurement evolution and repeated records, while a separate effective spatial model executes state-field feedback.

Outputs. The connected construction yields the uncalibrated finite residue 4.876893353%, joint address calibration 5% / 26.8% / 68.2%, and baseline conditional excitation 34.4764357391 MeV. The two named energy calibrations reproduce present deceleration -0.52855 and -0.523 . Ideal measurement and recording work is linked to finite supply boundaries; the declared record-noise model gives error 0.2005056913 after 256 ticks. This execution passes 1,116 bridge stage checks and 188 review checks, and reproduces 96 distinct downstream check groups and 37 result/source files.

Falsifiers. A mismatch in normalization, positivity, branch completeness, energy balance, pressure derivative, currents, time phase, input dependence or convergence rejects or revises the corresponding transformation. Observations inconsistent with a conditional output under matched calibration and physical preparation require reconsideration of that rule.

Conclusion. We integrate finite generation, common-state physical readout, measurement and record costs, and geometric response through executable connections using minimum computation, a common carrier and phenotype readout. The result establishes reproduction, explanation and conditional predictions for an adopted construction of one nonprivileged universe. Autonomous supply and measurement apparatus, actual species abundance, physical clock-rate identification and a common SI covariant evolution of homogeneous background and nonuniform states remain explicit interfaces.

Keywords WRRM minimum computation common carrier two stage filter phenotype information load energy ledger quantum measurement proper time

1 Research objective and meaning of integration

WRRM aims to explain one universe without privileged status through minimum computation, a common carrier and phenotype readout. WRRM specializes that structure to an adopted our-universe execution using established constants and observations. Minimum computation means a sufficient organization of reusable information and operations, including necessary reserve and robustness. We select one possible construction without asserting a unique universe or a global optimality theorem.

Upstream studies construct what is generated, admitted and retained and how a phenotype is selected. Downstream studies calculate the energy, pressure, currents, clocks and geometry read from a supplied state. Connecting them requires actual transfer of normalization, state, units, boundary and time. Shared symbols and equal totals alone do not establish an implemented connection.

Verified inputs → WRRM transformation → outputs → falsifiers. We take upstream integrated 1.0 r1 [1], downstream integrated 1.0 and its executable series [2], and reviewed bridge collections [3,4]. We trace numerical dependencies to distinguish executed transfers from retained provenance. Address-conditioned states connect to the SI ledger and homogeneous expansion; the same internal generator connects to measurement and repeated readout. If perturbing an allegedly connected input does not propagate through its declared calculation, that integration claim fails.

Verdict. Originality is assessed in the address construction, ordered filters, conditional preparation, common energy-current readout and boundary accounting. Numerical novelty is not a prerequisite. Model revision using verified values and their reproduction are legitimate consistency and explanatory achievements. An unmeasured physical quantity calculated after fixing verified inputs and construction choices is a conditional WRRM prediction. A matching number in another theory does not remove attribution to the independently executed WRRM transformation. External equations count as integrated only where executed within the stated ledger, and the same evidentiary standard applies to every model.

2 State spaces and calibration ledger

Addresses, carrier support, particle internal states and channel representations are different spaces. An address is not automatically a species, physical length or bit count. Information load is a positive state-dependent operator expectation, rather than Shannon information or channel count. Phenotype denotes a selected readout; sectors without that readout may retain positive load.

Table 1 State spaces and scopes

Object	Definition in this paper	Distinct object
Finite addresses	$2 \leq n \leq 1,000,000$	Species or spatial coordinates
Reference address 1	Empty prime product and vacuum reference	Automatic physical vacuum cancellation
Internal state	Two selected modes in a symmetric 95D nucleon space	128D carrier support
Carrier	Stationary I/128 and adopted spatial transport	15 or 16 channel inventory

Object	Definition in this paper	Distinct object
Channels	15-channel structure and conditional 16-channel extension	Complete generation masses and mixing
Resident Actual	Phenotype ϕ plus nonphenotypic D	Complete ledger including return R
Physical record	Outcome-coupled finite register	Calculation log or provenance file

Table 2 Fixed inputs and construction choices

Item	Value or choice	Role
c	299792458 m s ⁻¹	Exact SI reference
h	6.62607015 $\times 10^{-34}$ J s	Exact SI reference
1 eV	1.602176634 $\times 10^{-19}$ J	Exact conversion
G	6.67430 $\times 10^{-11}$ m ³ kg ⁻¹ s ⁻²	Fixed gravity calibration
H ₀	67.4 km s ⁻¹ Mpc ⁻¹	Adopted homogeneous density benchmark
Electron rest energy	510998.95069 eV	Electron-mode calibration
Address targets ϕ D R	0.05 / 0.268 / 0.682	Joint address calibration
Inherited energy targets ϕ D R	0.0493 / 0.265 / 0.6857	Inherited SI-load calibration
Alternate energy targets ϕ D R	0.05 / 0.268 / 0.682	Separately declared SI calibration
Reference volume V ₀	1 m ³	Ledger reference volume
Source	Four zeta-zero heights and eight frames	Phase-driver construction
Preparation strength	a _p = 0.2	Conditional excited population
Electron mode	23 and zero intercept	Adopted energy unit
Supply reservoir	0.2 E ϕ ,ref	Finite exchange diagnostic
Record noise	pflip = 0, 0.001, 0.01	Finite-record inputs

The exact SI constants have the status defined by BIPM [10]. G and electron energy are fixed verified inputs and are not counted as new independent predictions. H₀ is an adopted Planck-based model calibration. The rounded 5% / 26.8% / 68.2% triplet is not presented as an exact Planck parameter triplet [11]. Particle masses, axial charge, magnetic moments, charge radii, excitation centre and beta-decay normalization retain the upstream input ledger [1,5,6].

Verified inputs → WRRRA transformation → outputs → falsifiers. We distinguish verified values, unit definitions and construction choices and preserve separate state-space types. Physical response coefficients are declared when address fractions become energy fractions. Resident 31.8%, complete 100%, and address versus energy fractions then have consistent meanings. Direct addition across different normalizations or substitution across different physical types is rejected.

Verdict. Integration rests on transfers that preserve type and unit. Generation/channel allocation, internal mode support and spatial grid size are separate specifications.

3 Common carrier and conditional channel selection

The downstream structure starts with two seven-dimensional sectors and one invariant channel. Under the $SU(3)$ colour subgroup each seven branches into a triplet, antitriplet and singlet. The fifteen channels therefore correspond to the colour multiplicities of one minimal Standard Model generation [7]. Seven does not denote measured spatial directions.

$$C_{15} = 1_0 \oplus 7_A \oplus 7_B, \\ 7 \downarrow SU(3) = 3 \oplus \bar{3} \oplus 1. (1)$$

The transported neutral extension gives sixteen channels with adopted pairing (4,2,1) and (anti-4,1,2). Conditional on this pairing one hypercharge rule returns the known one-generation charges and anomaly cancellations.

$$C_{16} = C_{15} \oplus 1_N, \\ Y = T_{3R} + \frac{B-L}{2}, Q = T_{3L} + Y. (2)$$

Compatibility among four colour-preserving filters is the negative squared response mismatch in a common positive carrier metric. The target is the unique maximum when both left and right score gaps are positive. Positive-rate replicator selection converges when initial winner support is nonzero; entrywise perturbations of size ρ preserve selection when the smaller gap exceeds 4ρ .

Verified inputs → WRR transformation → outputs → falsifiers. Fixed pairing and origin-calibrated responses enter the common-metric scores. Uniform-reference responses 0.672634137 and 0.825428002 yield left/right gaps 0.04669193039, and the target wins nine disclosed state-volume cases. Loss of positive gaps, transport rank or winner support invalidates the corresponding unique-selection claim.

Verdict. Common transport and conditional channel selection execute. The symmetric prototype's zero gaps remain a failed selection example. The origin of calibrated orientation, Yukawa interactions, generation mixing and the complete interacting Standard Model are outside this calculation. Gauge neutrality does not determine an observed sterile-neutrino mass or population.

4 Finite SOURCE and ordered two stage filtering

Finite address weights reuse prime-factor ancestry. A prime is an independent SOURCE address; a composite is a relation retaining factors and powers. The fold index $\Omega(n)-1$ is an arithmetic organization index rather than a measured spatial dimension. A composite admitted under blurred prime-like recognition remains arithmetically composite.

$$Z_N(\alpha) = \sum_{n=2}^N n^{-\alpha}, \\ w_n = \frac{n^{-\alpha}}{Z_N(\alpha)}, \sum_n w_n = 1. (3)$$

The instantiated calculation is the finite sum. Euler products ground prime-factor reuse mathematically [9]. An uninstantiated infinite physical resource is Null in the execution ledger. Mathematical limiting comparisons do not require physically realized infinity, and the finite construction introduces no such resource.

Admission A and return B are contractions satisfying $A \dagger A \leq I$ and $B \dagger B \leq I$. Retention and complete return are

$$q_{res} = \text{Tr}[\rho A^\dagger (I - B^\dagger B) A],$$

$$q_R = 1 - \text{Tr}(A \rho A^\dagger) + \text{Tr}(B A \rho A^\dagger B^\dagger). \quad (4)$$

Their sum is one. General operators depend on order. In the diagonal implementation, normal return selects primes, even composites occupy D, and selected odd-composite admission occupies ϕ . Constant admission β gives

$$E_\phi = \beta P_{odd,c}, E_D = P_{even,c},$$

$$E_R = P_{prime} + (1 - \beta) P_{odd,c}, \sum_\ell E_\ell = I. \quad (5)$$

Phenotype readout pairs $n = 2m+r$ through the isometry $V|n\rangle = |m\rangle \otimes |r\rangle$ and combines the unpaired $r = 1$ port with composite selection. Initial reflection and normal return are distinct Kraus branches; their coarse R state is the sum of densities, not an arbitrarily summed pure amplitude. Persistent retention additionally requires subsequent updates to preserve $\ker B$.

Table 3 Finite composition with one denominator

Case	ϕ fraction	D fraction	R fraction
$\alpha = 2 \beta = 1 \ N = 10^6$	4.876893353%	24.999961236%	70.123145410%
$\alpha = 1.8996876951 \ \beta = 0.8654570124$	5.000000%	26.800000%	68.200000%

Verified inputs $\rightarrow$ WRR transformation $\rightarrow$ outputs $\rightarrow$ falsifiers. Finite support, exponent and admission rules produce positive effects under one normalization. The fixed-exponent uncalibrated residue and two-parameter calibrated composition are separate outputs. R follows from completeness in the calibrated row. Negative effects, an incomplete branch sum, duplicate returns or unrecorded retained-subspace leakage reject the construction.

Verdict. The model reproduces the adopted composition through its filter structure and separately calculates the fixed-exponent near-5% residue. The 4.8769% and 26.7942% comparisons at different exponents cannot be added directly. Address reproduction and physical-response calibration are both legitimate, separate construction operations.

5 Frame admission and shutter recording

The driver uses four zeta-zero heights γ_j , equal coefficients and phase increment $\xi = 0.1$. Admission consumes only pending addresses, preventing repeated counting [1,2].

$$\delta_n(k) = \sum_j c_j \cos[\gamma_j (\log n + \xi k) + \theta_j],$$

$$a_n(k) = \frac{1}{1 + \exp[h - \delta_n(k)]}. \quad (6)$$

$$T_n(K) = 1 - \prod_{k=0}^{K-1} [1 - a_n(k)],$$

$$\phi + D + S + R = 1. \quad (7)$$

Pending S enters R at the declared recovery boundary. At $K = 8$, $h = 1.447673170352440$ returns 5%. Holding α and h fixed and closing at four or sixteen frames gives $\phi = 3.704795\%$ or 5.682474% . Separately refitted drivers can reproduce 5% while retaining different conditional address distributions. Equal totals do not identify the same generation history.

The coarse φ D R shutter is a completely positive trace-preserving channel that retains branch probabilities and reduces coherence. Record strength η_{rec} is distinct from magnetic calibration η .

$$S_{\eta_{rec}}(\rho) = (1 - \eta_{rec})\rho + \eta_{rec} \sum_{\ell} P_{\ell} \rho P_{\ell}, 0 \leq \eta_{rec} \leq 1. \quad (8)$$

Verified inputs $\rightarrow$ WRR transformation $\rightarrow$ outputs $\rightarrow$ falsifiers. Baseline branches, shutter and subsequent mixing determine an order comparison. Full-record purity is 0.539448. At mixing angle 0.25, rotate-then-record gives $\varphi = 0.007845982335$ and record-then-rotate gives 0.063343500754. A shutter-only probability change or loss of positivity/trace rejects the channel.

Verdict. Recording can preserve immediate probabilities while altering subsequent interference. The coarse upstream shutter and the finite physical register with an explicit energy allocation are different stages. The selected zeta driver is not required to be the unique cosmic driver.

6 Stable internal states and common energy current readout

The particle example selects residue markers 45 and 75 and prime families 3 and 5. Proton and neutron species are externally assigned. The initial quadratic spatial coupling returns axial and magnetic anchors in a two-mode block but fails a lower-bound test under spatial enlargement. We preserve that failure and adopt a bounded interaction [5].

$$\begin{aligned} W_{\lambda} &= \frac{W_0}{k_{\lambda}(1 + \lambda s)}, \\ s &= \frac{x^2 + y^2}{\ell^2}, \lambda > 0, \\ H_B &= (E_{B,rest} - \Delta e_0)I + \Delta h. \quad (9) \end{aligned}$$

Positive kinetic-confining terms with bounded coupling supply a lower bound. $K = 8$ has 165 spatial modes and 95 symmetric states. Axial calibration gives $x_c = 0.8208877164$. Assigning the first positive-parity excitation centre to 1370 MeV gives $\Delta = 507.032217686$ MeV and $\text{gap} = 431.081244315$ MeV. The discrete Hermitian model does not calculate a resonance width or complex pole.

The original point-charge model gives neutron mean-square radius -0.186912 fm^2 , disagreeing with -0.1155 fm^2 . A zero-charge isovector term and common width jointly recalibrate both charge radii, yielding $\ell = 0.65715059784 \text{ fm}$ and $cE = -0.09708290347 \text{ fm}^2$ [6]. This is verified-input model construction; the dimensionless internal Hamiltonian and excitation scale remain fixed.

Energy, charge, magnetic and axial currents are evaluated in the same state. Magnetic response is the field derivative of the same Hamiltonian. The actual finite-momentum $n \rightarrow p$ current agrees with the electromagnetic isovector difference, and projected longitudinal current satisfies internal transition continuity. A complete transverse relativistic gauge current has not been executed.

Verified inputs $\rightarrow$ WRR transformation $\rightarrow$ outputs $\rightarrow$ falsifiers. Particle anchors and declared spatial/current responses enter the common 95D state. At $Q^2 = 0.1 \text{ GeV}^2$, ground readouts are $G_E p = 0.7367799$, $G_E n = 0.0399134$, $M_p = 1.9543552 \text{ } \mu\text{N}$, $M_n = -1.4207472 \text{ } \mu\text{N}$ and $G_A = 0.9039833$. Failure of Hermiticity, symmetry, field derivative, current continuity or spatial convergence rejects that internal construction.

Verdict. Common-state mass, excitation and current readout execute. The uncalibrated neutron magnetic radius 0.831987107 fm versus 0.864 fm remains a discrepancy. Beta decay also uses the common currents and phase space, but fixed- κ lifetime 846.204102380 s and refitted- κ return 878.3 s remain separate ledgers. Higher radiation-recoil terms, a microscopic γW box and induced-pseudoscalar completion are outside the calculation [6].

7 Address conditioned internal state and common carrier

Conditional preparation makes address residues influence internal energy and currents. $\chi(n)$ marks odd prime-3 valuation $\nu_3(n)$. For phenotype-conditioned probability $p(n|\phi)$, g and e_1 are frozen ground and first-excited modes. Preparation strength a_p is distinct from cosmic scale factor a .

$$\rho_{int}(n) = [1 - a_p \chi(n)] |g\rangle\langle g| + a_p \chi(n) |e_1\rangle\langle e_1|,$$

$$p_{exc} = a_p \sum_n p(n|\phi) \chi(n). \quad (10)$$

$$\Sigma_\phi = \sum_n p(n|\phi) |n\rangle\langle n| \otimes \rho_{int}(n) \otimes \frac{I_{128}}{128}. \quad (11)$$

This is a diagonal-address classical-quantum state. Exact block contractions avoid allocating a dense million-address tensor matrix. Conditional trace is one and complete-ledger trace of $f\phi\Sigma\phi$ is $f\phi$. Nucleon preparation is not arbitrarily assigned to D or R. Preparation-map Kraus completeness gives positivity and normalization.

Table 4 Baseline readouts from the prepared common state

Quantity	Value	Unit or scope
Selector fraction t	0.3998832725128899	Conditional on ϕ
Excited population p_{exc}	0.07997665450257799	$a_p = 0.2$
Conditional excitation	34.47643573911998	MeV
GE p / GE n	0.730176767 / 0.041662690	$Q^2 = 0.1 \text{ GeV}^2$
M p / M n	1.935186410 / -1.409366748	μN
GE V / GA	0.688514077 / 0.897125841	$Q^2 = 0.1 \text{ GeV}^2$
Address-weighted energy difference	0.1447722277	MeV diagnostic

Address-independent operators have the same expectation in the joint state and product of marginals. An address-sensitive response $r(n) = 1 + 0.25 \log(n)/\log(N)$ diagnoses conditional correlation.

$$E_{joint} - E_{product} = a_p g a p (\langle r \chi \rangle - \langle r \rangle \langle \chi \rangle). \quad (12)$$

Verified inputs → WRR transformation → outputs → falsifiers. We combine the same address state and frozen internal modes in conditional blocks. Baseline excitation and currents share one density, and the correlation diagnostic equals Eq. 12. Inconsistent block trace, positivity, marginal current or covariance rejects the bridge.

Verdict. The connection extends the former address-carrier product with explicit address-internal correlation. It does not establish general entanglement or coherent address preservation. The 0.1447722277 MeV diagnostic is not an additional cosmic energy term. A

phase-only $\theta_3 = 0.7$ control leaves the dephased prepared population unchanged, as required.

8 Nonduplicated SI energy and supply boundaries

Address probabilities do not automatically determine particle number. In pure proton or neutron conditional examples, expected population $\bar{N}$ at V_0 matches the existing phenotype energy allocation. $\bar{N}$ below one is an expected occupancy rather than production of a fractional particle in an individual run.

$$\bar{N} = \frac{E_{\phi,ref}}{E_{B,rest} + E_{exc,base}},$$

$$E_{sys} = \bar{N}(E_{B,rest} + E_{exc}) + E_D + E_R. \quad (13)$$

Equation 13 replaces E_ϕ by its rest/excitation decomposition. Adding that decomposition to the unchanged total would duplicate all of E_ϕ . The inherited proton example has $\bar{N} = 0.2425893464$, with rest $3.6467913480 \times 10^{-11}$ J and baseline excitation $1.3399990150 \times 10^{-12}$ J. The conversion is $1 \text{ MeV} = 1.602176634 \times 10^{-13}$ J.

$$\Delta E_{sys} = \bar{N}(E_{exc,case} - E_{exc,base}),$$

$$\Delta E_{env} = -\Delta E_{sys}. \quad (14)$$

With the supplier outside the boundary, system increase is external work. Including the supplier cancels both increments and conserves total energy. Supplier state and pressure need separate specification. The diagnostic uses initial reservoir $0.2 E_{\phi,ref}$ and pressureless fixed comoving energy; it does not introduce a new cosmic fraction.

Verified inputs → WRR transformation → outputs → falsifiers. The same excitation, MeV conversion, fixed expected population and finite reservoir determine replacement and signed exchange. A small preparation variation supplies $8.9385339569 \times 10^{-15}$ J in the proton example and an opposite environmental increment. Negative reservoir, insufficient supply, nonfinite data or changing included total cause rejection.

Verdict. Conditional excitation differences enter a physical SI ledger with explicit boundary meaning. Downstream admission conversion work $1.3279304601 \times 10^{-12}$ J is not added to internal preparation work without identifying whether they represent the same event. A joint autonomous source-driver-apparatus-reservoir Hamiltonian has not been executed.

9 Pressure expansion and local gravity from one energy

Positive downstream carrier loads are $A_\phi = I$, $AD = Kc/2$ and $AR = Kb/2$. Kc is the positive semidefinite discrete negative Laplacian on the periodic carrier; Kb is the declared noncommuting positive control. Address responses μ_ℓ and SI coefficients η_ℓ define one energy. During the fixed-state, fixed-population post-preparation interval, ϕ and D have constant comoving energy and R has constant density.

$$u_{crit} = \frac{3H_0^2 c^2}{8\pi G},$$

$$E_\ell = V_0 \eta_\ell \mu_\ell a^{v_\ell} \text{Tr}(\rho A_\ell),$$

$$(v_\phi, v_D, v_R) = (0, 0, 3). \quad (15)$$

$$V = V_0 a^3, u = \frac{E}{V},$$

$$P_\ell = -\partial_V E_\ell = -\frac{v_\ell}{3} u_\ell. (16)$$

The homogeneous bridge has $E(V) = M + LV$, $P = -L$ and $u = M/V + L$. The adopted homogeneous gravity response acts on the same energy and pressure.

$$H^2 = \frac{8\pi G}{3c^2} u,$$

$$q = \frac{u+3P}{2u}, a \frac{du}{da} + 3(u+P) = 0. (17)$$

Global weighted twist reads D+R, while local extra attraction reads D. Stress energy and its geometric readout are counted once. The local response and factor 1/8 are declared constitutive choices.

$$a_T = c H_0 \sqrt{\frac{u_D}{8u_{crit}}},$$

$$\zeta_T \kappa_h^2 = \frac{16\pi G}{c^4} (u_D + u_R),$$

$$g = \frac{g_M}{1 - \exp[-\sqrt{g_M/a_T}]}. (18)$$

Table 5 Two named energy calibrations with the same rules

Output	Inherited calibration	Alternate calibration
Energy fractions ϕ D R	4.93% / 26.5% / 68.57%	5% / 26.8% / 68.2%
Reference density $J\ m^{-3}$	$7.6689477678 \times 10^{-10}$	$7.6689477678 \times 10^{-10}$
Reference pressure Pa	$-5.2585974844 \times 10^{-10}$	$-5.2302223777 \times 10^{-10}$
q_0	-0.528550000	-0.523000000
Trial rotation $km\ s^{-1}$	207.510905127	207.688109769
Conditional deflection arcsec	0.535586511	0.536727827

Rotation is the inherited downstream output at 8.2 kpc for a Plummer trial source of 6×10^{10} solar masses and scale 3 kpc. Lensing uses impact parameter 10 kpc, boundary 200 kpc and $\Phi = \Psi$. These are not new fits to observed galaxies. The present bridge calculation concerns the shared ledger and homogeneous replay; rotation/lensing values cite the replayed downstream source results.

Verified inputs → WRRRA transformation → outputs → falsifiers. Each named calibration and boundary is evaluated through the same $E(V)$ derivative and homogeneous integration. Both q_0 values and $H/H_0 = 1$ reproduce, with maximum scale-factor error 3.297429×10^{-11} across 32 integrations against the analytical solution. Failed pressure derivative, continuity, independent acceleration or nonduplication rejects that connection.

Verdict. Gravity connects the declared information load to physical energy, pressure and geometry. The explanatory phrase weight of information denotes this transformation, not an uncalibrated conversion of address norm into kg or J. q is an output of fixed component pressures, not its own fitting target. Included-supplier $q_0 = -0.5185075159$ or -0.5128712871 describes a different boundary. A common $C(V)I$ added to normalized active

and vacuum-reference states cancels in their relative energy and derivative; state-specific energy changes and R background remain [8]. This diagnostic does not replace covariant interacting quantum-field vacuum dynamics.

10 Proper time and conditional SI frames

Electron mode 23 gives $\mu E = E_e/23 = 22217.345682174$ eV and $t_\mu = \hbar/\mu E = 2.962603932832 \times 10^{-20}$ s. The downstream update $0.05t_\mu$ differs from the source frame assigned to phase increment 0.1.

$$\begin{aligned}\omega &= \frac{\mu E}{\hbar}, \\ \Delta\tau_{frame} &= \frac{0.1}{\omega} = 2.962603932832 \times 10^{-21} \text{ s}, \\ \Delta\tau_{down} &= 0.05t_\mu. \quad (19)\end{aligned}$$

Bridge 0.5 defines a five-dimensional source generator $H_{src} = \text{diag}(0, \mu E\gamma_1, \dots, \mu E\gamma_4)$ and address-phase states. Direct matrix exponentiation and readout reproduce the frame driver $0.5\Sigma_j \cos[\gamma_j(\log n + \omega\tau)]$. The autonomous coupling that turns this readout into sigmoid admission remains unconstructed.

Using the same proper time in the 95D internal generator gives first-gap phase approximately 1940.2913853 rad per frame. Ninety-four of the 95 relative energies require nonzero integer branches. Reading phase alone and discarding branches loses the original energy. The executed relative generator is $H_{rel} = H_B - E_{B,rest}$. Its eigenvalues $\epsilon_j = E_j - E_{B,rest}$ are expressed in joules in Eq. 20; $\tau = k\Delta\tau_{frame}$. Here θ_j is the principal phase of the corresponding eigenvalue of $U_{rel}^\dagger$, matching the positive generator-phase convention.

$$\begin{aligned}U_{rel}(\tau) &= e^{-iH_{rel}\tau/\hbar}, \\ \epsilon_j &= \frac{\hbar}{\tau}(\theta_j + 2\pi b_j), b_j \in \mathbb{Z}. \quad (20)\end{aligned}$$

Updates attach to each timelike worldline's accumulated proper time. At coordinate duration 64 frames, flat specified speeds $\beta = 0, 0.6$ and 0.8 give 64, 51 and 38 complete frames, with fractional remainders retained. A Lorentz control preserves the same proper time. No rest-clock count is assigned to null paths; that does not freeze light propagation.

Verified inputs → WRR transformation → outputs → falsifiers. Fixed electron energy, source phases and the same internal H are evolved at the adopted SI spacing. Twenty-five address-frame cases reproduce the driver, internal phases and energy recovery, with maximum recovery error $1.8189894035 \times 10^{-12}$ MeV. Lost phase branches, inconsistent units or failed proper-time invariance reject the clock connection.

Verdict. Conditional SI phase connection executes. Doubling driver rate while halving frame duration preserves frame unitaries, so the actual physical rate is not uniquely identified by dimensionless frame data. The old slow-schedule/SI-generator ratio $2.102556972196466 \times 10^{-43} \neq 1$ remains failed. The separately chosen generator does not repair that old equality. Adopted spacing and cavity lengths are not measured minimum time, minimum length or cosmic size.

11 Post measurement evolution and record work

Bridge 0.6 applies an X readout on the invariant two-mode support. The basis is $|\pm\rangle = (|g\rangle \pm |e_1\rangle)/\sqrt{2}$; the finite two-state record begins blank $|0\rangle$, with record gap μE converted to SI. The writing unitary and conditional state are

$$W = P_+ \otimes I_{rec} + P_- \otimes X_{rec},$$

$$\rho_{\pm} = \frac{P_{\pm} \rho P_{\pm}}{\text{Tr}(P_{\pm} \rho)}. \quad (21)$$

The diagonal baseline gives equal 0.5 outcome probabilities because of the selected basis. Readout acts only within ϕ , giving five route probabilities $\phi/2$, $\phi/2$, D , initial reflection and normal return. Coarse ϕ is preserved. Evolution with the same relative H followed by another X readout gives Eq. 22, where $\text{gapJ} = \text{gapMeV} \times 1.602176634 \times 10^{-13} \text{ J/MeV}$ and $\hbar$ is in J s. The cosine argument is dimensionless.

$$P_{same}(k) = \cos^2 \left(\frac{g a p_J k \Delta \tau_{frame}}{2 \hbar} \right),$$

$$\epsilon_{rec}(m) = \frac{1 - (1 - 2 p_{flip})^m}{2}. \quad (22)$$

W generally does not commute with system plus record energy, so externally controlled work is required. Mean single-particle writing work is $2.901146067898069 \times 10^{-11} \text{ J}$. The 0.8 review inserts that cost once using the reference expected population.

$$\overline{W} = (1/2 - p_{exc}) g a p_J + \frac{E_{rec, gap}}{2},$$

$$\Delta E_{supply} = \overline{N} \overline{W} = -\Delta E_{env}. \quad (23)$$

Table 6 One mean write and supplier boundaries

Case	Supply J	External q before and after	Included q
Inherited proton	$7.0378713 \times 10^{-12}$	$-0.528550 \rightarrow -0.519197$	-0.518508
Inherited neutron	$7.0285264 \times 10^{-12}$	$-0.528550 \rightarrow -0.519209$	-0.518508
Alternate proton	$7.1378005 \times 10^{-12}$	$-0.523000 \rightarrow -0.513566$	-0.512871
Alternate neutron	$7.1283229 \times 10^{-12}$	$-0.523000 \rightarrow -0.513579$	-0.512871

Inherited proton excitation rises $7.0374395222 \times 10^{-12} \text{ J}$ and record energy rises $4.3176187869 \times 10^{-16} \text{ J}$. The reservoir falls from $7.5615824991 \times 10^{-12} \text{ J}$ to positive $5.2371121503 \times 10^{-13} \text{ J}$. With the supplier included and the pressureless volume law retained, total energy and q remain unchanged by redistribution. The external and included boundaries begin with different totals and therefore have different q .

Verified inputs → WRR transformation → outputs → falsifiers. Frozen gap, population, spacing and record gap specify the finite readout. Born probabilities, conditional states, mean work, repeated transitions and record noise are calculated. At $p_{flip} = 0.001$, 256 ticks give error 0.200505691292 over the adopted time $7.5842660680 \times 10^{-19} \text{ s}$. Internal plus record increase differing from supply, a changing included total or a negative reservoir rejects the connection.

Verdict. Outcomes, post-observation states, repeated statistics and a finite record, previously absent from standalone downstream 1.0, are conditionally executed by the bridge. The 100,000 seeded samples are computational statistics rather than independent physical experiments or accumulated costs of 100,000 physical memories. Blank preparation, reset, thermal bath, long-term storage, autonomous control, full currents of coherent post-observation states and apparatus distribution remain open. One mean-cost insertion is not a time-resolved $Q(t)$ or apparatus dynamics.

12 Finite nonuniform state field feedback

Bridge 0.7 transfers baseline internal populations to two carrier components with different initial spatial profiles. They share transport and have no new intercomponent transitions. This post-preparation closed interval is a dimensionless nonrelativistic effective model.

$$\begin{aligned} i \partial_s \psi_j &= \left[-\frac{1}{2} \partial_x^2 + \Phi \right] \psi_j, \\ \partial_x^2 \Phi &= \lambda (\rho - \bar{\rho}), \\ \rho &= \sum_j |\psi_j|^2. \end{aligned} \quad (24)$$

Periodic Poisson subtracts mean density and fixes mean $\Phi = 0$. Fourier differentiation and a symmetric split step recalculate the field from each updated density. Occupancy-to-field-to-state feedback therefore actually executes. Self-consistent dynamic energy retains the half interaction factor.

$$E_{dyn} = \frac{1}{2} \int \sum_j |\partial_x \psi_j|^2 dx + \frac{1}{2} \int \rho \Phi dx. \quad (25)$$

Verified inputs → WRR transformation → outputs → falsifiers. With $L = 20$, $\lambda = 1$, periodic boundaries and Gaussian profiles, runs reach $s = 1$. At $N = 64$ and steps 0.01, 0.005 and 0.0025, successive density differences have ratio 4.000041. The $N = 64/128$ grid L2 difference is 9.02348×10^{-12} . Dynamic-energy drift is checked without diluting it by large constant rest/internal energy, alongside Poisson residuals, local current continuity, reversal and free controls. Failed normalization, currents, convergence or energy diagnostics reject this effective interval.

Verdict. Finite nonuniform self-consistent feedback executes. Its source is equal-weight carrier occupancy, not a stress-energy tensor including excitation energy. The 0.8 dependency probe preserves the finding that doubled clock rate or record work does not change its density/field. Changing population to 0.12 changes final density by up to 0.01681125416, confirming population as an actual numerical input while clock/record files are provenance. Uniform $\Phi = 0$ follows from mean-subtracted Poisson and does not reproduce FRW expansion. Common SI kinetic, length and G calibration, post-measurement state transfer to fields and covariant background/perturbation integration remain open.

13 Consistency explanation and predictions

Table 7 Apparent conflicts and their resolution

Issue	Integrated ledger resolution
Resident 31.8% versus complete 100%	Separate $\phi + D$ retention from the ledger including return
Address 5 26.8 68.2 versus inherited energy fractions	Distinguish probability measure and named SI response calibration
$q - 0.52855$ versus -0.523	Preserve different calibration outputs of the same pressure law
Rest/excitation versus existing E_ϕ	Replace the existing allocation rather than add it twice
Post-readout $p_{exc} = 0.5$ versus fixed prepared state	Supply writing work and define a distinct post-write interval

Issue	Integrated ledger resolution
Shutter versus register record	Distinguish coarse channel from physical energy and medium
Shared time versus distinct generators	Separate 5D source and 95D internal H
Uniform Poisson $\Phi = 0$ versus expansion	Separate mean-free spatial response and FRW background
Clock/record files versus field integration	Decide integration through actual input dependencies
Infinite comparison versus finite realization	Execute finite sums and use limits as mathematical comparisons

Verified inputs → WRR transformation → outputs → falsifiers. We label scope, state, unit, calibration and boundary before comparing outcomes. These ledger and physical issues admit consistent simultaneous description while preserving actual address-state-SI-measurement-geometry achievements. A disagreement at the same inputs and boundary, or a claim of execution for an unimplemented edge, requires correcting the synthesis.

Verdict. The present architecture resolves conflicts produced by forced identification of distinct ledgers. This does not complete actual abundance or a full SI covariant simulator. Missing connections remain explicit calculations to perform without erasing the reproduced results.

Conditional WRR predictions after fixing inputs include source-dependent excitation and currents, repeated transitions at the same gap/frame, noise-conditioned record errors and specified spatial density/field evolution. Their physical tests must match the preparation and observable. Doubling the adopted frame changes the first repeated-transition matrix by a maximum 0.55243420946, demonstrating that the time adapter actually affects repeated readout.

Established constants, particle properties and composition returns are reproduction and explanation. Trial-source rotation/lensing are conditional calculated outputs. Unmeasured prepared-state responses under fixed physical mappings are conditional WRR predictions. Code comparisons and seeded samples are implementation evidence. These contributions are not dismissed solely because numbers are known, and different evidence types are not combined into a single proof count.

14 Reproduction and integrated conclusion

This synthesis introduces no new observational fit. Both energy calibrations are retained while state, measurement and supply transfers are replayed. The reviewed bridge runs fresh stages 0.1 through 0.7 in dependency order and then the 0.8 review. Counts 15, 95, 134, 600, 134, 87 and 51 total 1,116 stage checks; 188 review checks give 1,304. They include mathematical, implementation, provenance and expected-failure checks, not observational experiments.

Downstream 0.10 through 0.12 were regenerated in a clean copy after removing result directories and bilingual computational manuscripts. Distinct group counts 32, 28 and 36 passed, and 37 result/source files matched the baseline byte for byte. The current archive manifest verifies 312 entries. Repeated dependencies are not added again as independent evidence. Upstream collection counts 395 for 0.4 through 0.6 and 1,395 for 0.7 through 0.10 are prior published records rather than new experiments in this manuscript.

Verified inputs → WRR transformation → outputs → falsifiers. Public fixed inputs, code, outputs and hashes are replayed in a clean environment. We check result identity,

dependency hashes and declared failures. State transfer, units, pressure and measurement checks pass here. Any manuscript-execution-hash disagreement requires correcting the publication specification.

Integrated conclusion. Finite upstream generation prepares a selected internal state whose energy and currents are read together. Its energy replaces the existing phenotype allocation without duplication and transfers through supply/record increments to common pressure and homogeneous expansion. The same internal generator and adopted proper time specify post-observation evolution and repeated probabilities. A separate effective spatial interval executes baseline-population state-field feedback. These actual connections constitute the integrated achievement.

Upstream closes at 0.10, downstream at 0.12 and bridge development at 0.8. Integrated 1.0 is one manuscript of the reviewed executable scope, not an automatic extension of development stages. Autonomous source-admission-preparation-measurement supply, species selection and actual abundance, durable records, physical clock rate and common SI covariant geometry remain declared research interfaces. The objective of explaining one nonprivileged universe through one method and the present computational reach are stated together.

Appendix A Executed paths and remaining interfaces

Path	Status	Actual transfer or missing calculation
Address → internal state → SI ledger → homogeneous expansion	Executed	External species/population and declared volume laws
Same internal H and SI frame → measurement/repetition	Executed	Finite controlled readout and conditional transitions
Baseline internal population → spatial density/field	Executed	Dimensionless nonrelativistic feedback
Mean write work → SI supply ledger	Review diagnostic	One mean insertion and pressureless included supplier
Clock/record → spatial field	Provenance only	No numerical dependency
Source → autonomous preparation/measurement supplier	Open	Joint Hamiltonian and actual time-resolved exchange
Background plus nonuniform state → common geometry	Open	Common-SI stress-energy tensor and metric evolution

Appendix B Reproduction commands and data

Extract the reviewed bridge r1 archive and run from its root.

```
OPENBLAS_NUM_THREADS=1 python reproduce_all.py
```

From the downstream 0.12 r1 archive root run

```
python calculations/wrra_m_0_10/run_release.py
python calculations/wrra_m_0_12/run_release.py
python review/0_10_to_0_12_r1/review_contracts.py
```

Python, numpy, scipy and mpmath follow the public requirements. The integrated distribution retains bilingual manuscripts, a numerical-claim/source-path register, a portable audit that reads the actual Word tables, fresh replay evidence, generation code, source archives and SHA256SUMS. Run `python audit_release.py` from the extracted package root to

check the manuscript values and source hashes. Existing document-review receipts are not used as validation of the new Word/PDF renderings.

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Authors and data availability

Wonsik Choi and Jeongin Choi contributed to the conceptual framework represented in the prior Physics preparation. Wonsik Choi developed and organized the calculations and manuscript materials. This attribution preserves the prior integrated manuscripts' declaration. The correspondence and ORCID belong to Wonsik Choi.

ChatGPT assisted mathematical presentation, code inspection and execution, synthesis, translation and document preparation. The authors retain responsibility for scientific claims and the final manuscript. This author-reviewed r1 integrated edition is identified by DOI <https://doi.org/10.5281/zenodo.23126800>. Source publications retain their distinct DOIs. The review corrects unit and relative-energy notation and strengthens the distribution audit; it introduces no new fitted physical parameters. No new journal submission or acceptance status is claimed.

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